



Python Assignment 02: Loops Practice

Objective:

Practice using **for loops** and **while loops** in Python to perform calculations and repetitive tasks.



Q1. Print Even Numbers between 1 and 20

Write a program to display all **even numbers** between 1 and 20 (inclusive).

Hint: Use the modulus operator `%` to check if a number is even.

Expected Output:

```
2 4 6 8 10 12 14 16 18 20
```



Q2. Sum of Natural Numbers

Write a program to calculate the **sum of the first N natural numbers**, where **N** is entered by the user.

Example Input/Output:

```
Enter N: 5  
Sum = 15
```

Explanation: $1 + 2 + 3 + 4 + 5 = 15$



Q3. Factorial of a Number

Write a program to calculate the **factorial** of a given number using a loop.

Formula: $n! = n \times (n-1) \times (n-2) \times \dots \times 1$

Example Input/Output:

```
Enter a number: 5  
Factorial = 120
```



Q4. Sum of Even and Odd Numbers

Write a program to calculate the **sum of even** and **sum of odd** numbers separately from 1 to N.

Example Input/Output:

```
Enter N: 10
Sum of even numbers = 30
Sum of odd numbers = 25
```

✓ Expected Learning Outcomes

After completing this assignment, students will:

- Understand how to **use loops** in Python.
- Apply loops for **counting, summing, and performing repetitive calculations**.
- Strengthen logical thinking with number-based problems.